From Dim Futures to Bright Horizons? Rural Electrification and Women's Empowerment in India

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September 2026

Abstract

Despite substantial global investments in rural electrification, access to electricity remains limited in many rural areas of developing countries. This paper examines the effects of India's Rajiv Gandhi Grameen Vidyutikaran Yojana (RGGVY), a large-scale rural electrification program launched in 2005, on women's empowerment. I employ a difference-in-differences instrumental variable (DID-IV) strategy that exploits predetermined program eligibility to identify variation in realized district-level electrification. The results show that electrification significantly reduces women's mobility restrictions and their acceptance of justifications for domestic violence while increasing intrahousehold agency, but has no statistically significant effect on financial autonomy. I examine three potential mechanisms underlying these findings: reduced domestic time burdens through changes in energy use and appliance adoption, greater access to information, and improved perceived safety. These findings suggest that rural electrification can strengthen women's position within the household and improve important dimensions of empowerment beyond the direct benefits of electricity access.

JEL Classification: O13, J16, O15

Keywords: rural electrification; women's empowerment; RGGVY

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1. Introduction

Can access to electricity change the balance of power within households? The United Nations' Sustainable Development Goals recognize universal access to affordable, reliable, and modern energy (SDG 7) and gender equality (SDG 5) as closely interconnected priorities (United Nations, 2015). Yet, as of 2018, approximately 800 million people worldwide lacked electricity, primarily in rural areas of low- and middle-income countries (Tracking SDG 7, 2020). The consequences of limited access are particularly important for women, who bear much of the responsibility for time-intensive domestic activities, including collecting fuel and water, cooking, and childcare (Ferrant and Thim, 2019; Standal and Winther, 2016). Identifying the causal effects of electrification on women's empowerment, however, remains difficult. Governments often implement large-scale infrastructure programs in the same areas to exploit potential complementarities across investments (Sanchez et al., 2007; Moneke, 2020), while the geographic concentration of such investments may also exacerbate spatial inequalities (Kanbur and Venables, 2005). When electrification coincides with other public investments, its effects are difficult to distinguish from those of concurrent interventions, particularly for outcomes such as women's empowerment, which evolve slowly and are shaped by deeply rooted social norms. Identifying the effects of electrification therefore requires disentangling its contribution from concurrent public investments and pre-existing socioeconomic differences across communities.

This paper examines the effects of a rural infrastructure policy in India that expanded access to electricity, focusing on women's autonomy and empowerment. While extensive research has assessed the effects of transport infrastructure on economic development (Ghani, Goswami, and Kerr, 2016; Asturias, García-Santana, and Ramos, 2019), poverty reduction (Khandker, Bakht, and Koolwal, 2009a), and employment (Roberts et al., 2018), a related literature has examined the effects of electricity infrastructure on employment, household income, asset ownership, and economic activity (Dinkelman, 2011; Grogan and Sadanand, 2013; Van de Walle et al., 2017; Lee, Miguel, and Wolfram, 2020; Burlig and Preonas, 2024). This study contributes to the literature by specifically examining the impact of rural electricity grid expansion on women's empowerment outcomes. Exploiting variation in electricity access induced by program eligibility, electrification reduces women's mobility restrictions, strengthens their

intrahousehold agency, and lowers their acceptance of justifications for domestic violence, while having no statistically significant effect on financial autonomy.

India, home to nearly 17% of the world's female population, offers a critical case for investigating this question. Despite progress in recent decades, India's female labor force participation remains low, hovering around 30%[1], and has been in decline and over 30% face restrictions on their mobility[2]. Such constraints on physical mobility are increasingly recognized as a key barrier to women's socio-economic participation (Bankar et al., 2018; Field & Vyborny, 2022). Societies with stringent mobility norms often also indicate lower labor force participation among women. In this context, studying how infrastructure interventions like rural electrification affect women's empowerment is important for both academic and policy purposes.

The expansion of the rural electricity grid took place under the Rajiv Gandhi Grameen Vidyutikaran Yojana (RGGVY), a flagship rural electrification program launched by the Government of India in 2005. The program aimed to provide electricity connections to over 100,000 unelectrified villages and improve electrification in more than 300,000 under-electrified villages across 27 states. India provides a particularly important setting for examining this expansion, having accounted for over 80% of the global increase in household grid connections between 2000 and 2016 (IEA, 2017). The scale of this expansion provides an opportunity to examine its socioeconomic consequences, particularly for women's autonomy and empowerment.

Rural electrification may improve women's empowerment through several interconnected mechanisms. First, access to modern energy and household appliances may reduce the time and effort women devote to domestic activities, creating opportunities for paid employment and other productive activities (Coen-Pirani et al., 2010; Dinkelman, 2011; Ferrant and Thim, 2019). Second, improved lighting may enhance women's perceived safety and facilitate greater mobility, allowing them to participate more freely in employment, education, and community activities (Clancy et al., 2002; Daka and Ballet, 2011). Third, electricity access may expand local economic opportunities and encourage the adoption of household technologies that facilitate

[1] This figure is based on the recent Periodic Labor Force Survey conducted in the year 2019–20.
[2] This figure is based on the India Human Development Survey conducted in the year 2011–12.

communication and home-based activities (Bose et al., 2022). Finally, increased access to television and other forms of mass media may expose women to alternative gender norms, potentially strengthening their intrahousehold agency and reducing their acceptance of justifications for domestic violence (Jensen and Oster, 2009; Winther et al., 2017). Together, these channels may improve women's bargaining position, although their effects may differ across dimensions of empowerment

The key empirical challenge in identifying the causal effect of rural electrification is that electricity infrastructure is not randomly allocated. Electrification decisions may reflect differences in economic development, infrastructure costs, governance, or political considerations, which may also independently influence women's empowerment (Khandker et al., 2014; Rathi and Vermaak, 2018; Sedai et al., 2021a). Consequently, comparisons between electrified and non-electrified districts may produce biased estimates. To address this concern, I exploit predetermined variation in RGGVY eligibility based on village population. Specifically, I instrument realized district-level electrification using an indicator for whether a district contained at least one previously unelectrified village that exceeded the program's population eligibility threshold of 300 persons. Since eligibility is determined using population recorded in the 2001 Census, four years before the program's launch, it provides a predetermined source of variation in electrification. Furthermore, Burlig and Preonas (2024) document a discontinuous increase in electrification around this threshold, providing evidence that the eligibility rule influenced subsequent electricity access.

This study uses data from two main sources. The primary dataset is the Indian Human Development Survey (IHDS), a nationally representative panel covering over 40,000 households across India. Conducted in 2004–05 and 2011–12, the survey provides detailed information on household energy use, socioeconomic characteristics, and women's autonomy. I use the eligible women's questionnaire to construct four measures of empowerment: mobility restrictions, intrahousehold agency, attitudes toward domestic violence, and financial autonomy. I merge the IHDS panel with village-level data from the 2001 and 2011 Population Censuses of India. Using these Census data, I identify initially unelectrified villages that achieved full grid electrification across the domestic, agricultural, and commercial sectors by 2011. I aggregate this information to the district level and merge it with the IHDS panel, restricting the analysis to rural women. I estimate

a difference-in-differences instrumental variable (DID-IV) specification with individual and survey-wave fixed effects, exploiting changes in women's empowerment over time. To assess the credibility of the identification strategy, I examine balance in predetermined district characteristics and conduct robustness checks controlling district- and village-level public goods provision.

The results show that rural electrification significantly improves several dimensions of women's empowerment. Electrification reduces women's mobility restrictions by 13.5 percentage points, increases their intrahousehold agency by 7.4 percentage points, and reduces their acceptance of justifications for domestic violence by 12.8 percentage points. These effects correspond to changes of approximately 19%, 10%, and 27%, respectively, relative to their pre-program means. In contrast, I find no statistically significant effect on financial autonomy, suggesting that improvements in women's mobility, household decision-making, and attitudes toward domestic violence do not necessarily translate into greater control over financial resources. The main findings remain broadly consistent across alternative measures of electrification exposure, an instrument based on villages near the eligibility threshold, and additional district- and village-level controls.

To explore the potential mechanisms underlying these findings, I examine changes in household energy use, appliance ownership, perceived safety, and access to information. The results show that electrification increases the use of electric lighting and is associated with greater use of modern cooking fuels, suggesting potential reductions in the time and effort required for domestic activities. Electrification also increases household ownership of electric fans, telephones, and sewing machines, which may improve living conditions and expand opportunities for home-based economic activities. Furthermore, I find a reduction in women's reported perceptions of harassment in their locality and an increase in regular television viewing, consistent with potential improvements in perceived safety and greater exposure to information and alternative gender norms. Together, these findings suggest several pathways through which electrification may influence women's empowerment

This paper contributes to a growing literature evaluating the effects of the Rajiv Gandhi Grameen Vidyutikaran Yojana (RGGVY), a large-scale rural electrification

program in India. Burlig and Preonas (2024) provide the first large-scale evaluation of the program and show that RGGVY substantially increased electricity access but had limited effects on a broad range of socioeconomic outcomes. More recent studies emphasize that the benefits of electrification may depend on complementary economic and infrastructure conditions (Fetter and Usmani, 2024; Vanden Eynde and Wren-Lewis, 2026). I contribute to this literature by examining the effects of RGGVY on women's empowerment. This study examines broader measures of women's empowerment, including mobility restrictions, intrahousehold agency, attitudes toward domestic violence, and financial autonomy. Together, these dimensions provide a more comprehensive understanding of how rural electrification can influence gender dynamics and women's position within households in rural settings.

Second, my findings complement Burlig and Preonas (2024), who show that RGGVY substantially increased electricity access but had limited effects on conventional measures of economic development, except in larger villages. In contrast, I find meaningful improvements in women's mobility, intrahousehold agency, and attitudes toward domestic violence. This suggests that evaluations focused only on conventional economic outcomes may overlook an important dimension of the welfare effects of electrification. At the same time, the absence of a statistically significant effect on women's financial autonomy is consistent with their finding of limited average economic impacts.

The remainder of the paper is organized as follows. Section 2 describes the institutional background of the RGGVY rural electrification program. Section 3 develops the theoretical framework linking electrification to women's empowerment. Section 4 presents the data sources, outcome measures, and sample construction. Section 5 outlines the empirical strategy and identification assumptions. Section 6 reports the main results, while Section 7 presents robustness checks and examines the potential mechanisms through which rural electrification affects women's empowerment. Section 8 concludes.

2. The RGGVY Electrification Program Context

Despite several rural electrification initiatives introduced since independence, a substantial proportion of villages and rural households in India remained without electricity in the early 2000s. To accelerate rural electrification, the central government launched the Rajiv Gandhi Grameen Vidyutikaran Yojana (RGGVY) in 2005, bringing existing grid-based rural electrification schemes under a single program. The program aimed to provide electricity connections to over 100,000 unelectrified villages (extensive margin) and improves electrification in over 300,000 under-electrified villages (intensive margin).[3] The program's goal was to strengthen the rural economy by expanding electricity infrastructure, including transmission and distribution lines and transformers, to support agricultural activities, educational and healthcare facilities, and small enterprises. It also provided free grid connections to below-poverty-line (BPL) households, while other rural households could obtain connections at their own expense.

RGGVY was largely funded by the central government and overseen by the Rural Electrification Corporation (REC), while implementation was delegated to state electricity utilities and other local implementing agencies. To obtain funding, implementing agencies prepared district-level Detailed Project Reports (DPRs), specifying infrastructure requirements, eligible villages, expected household connections, and estimated project costs. These reports were reviewed and approved by the REC and the Ministry of Power before funds were disbursed. Following approval, electrification proceeded according to the village-level implementation plans outlined in the DPRs. Projects were funded under India's 10th and 11th Five-Year Plans, with districts allocated according to the order in which their DPRs were submitted and approved. By 2011, the program had invested approximately Rs 253 billion (US$5.45 billion) and provided electricity connections to around 17.5 million rural households (Sreekumar and Dixit, 2011). An important feature of RGGVY was its population-based eligibility rule, under which previously unelectrified habitations with populations above 300 persons were prioritized for grid electrification (Burlig and Preonas, 2024). Since eligibility was

[3] By 2004, over 125000 villages lacked access to electricity and around 57% of the rural households in the remaining 467000 lacked the grid connection. The extensive margin refers to expanding electricity access to new users or areas previously without a connection, while the intensive margin focuses on improving the quality and reliability of power for those already connected.

determined at the habitation level, village population can be used to assess eligibility in single-habitation villages. This institutional feature is particularly relevant to the empirical analysis and is discussed further in Section 5.

3. Theory: Electrification and Women's Empowerment

3.1. Electrification and Intrahousehold Bargaining

The idea of empowerment emphasizes control over resources and the ability to make meaningful life choices. Kabeer (1999) conceptualizes empowerment through three interconnected themes: resources, agency, and achievements. For instance, resources refer to the material, human, and social preconditions that shape an individual's ability to exercise choice; agency captures the ability to define goals and act upon them; and achievements reflect the outcomes of those choices. This framework suggests that improvements in women's access to resources and capabilities can strengthen their ability to exercise agency and improve their well-being.

A large body of empirical literature measures women's empowerment using a multidimensional framework (Mahmud et al., 2012; Alkire et al., 2013). Following this approach, I focus on four dimensions of women's empowerment that are particularly relevant in the Indian context: mobility restrictions, intrahousehold agency, attitudes toward domestic violence, and financial autonomy. Together, these outcomes capture women's freedom of movement, participation in household decision-making, ability to challenge restrictive gender norms, and degree of control over economic resources.

To understand how electrification may affect these dimensions, I draw on the collective model of intra-household bargaining (Manser and Brown, 1980; Chiappori, 1988; Lundberg and Pollak, 1993). This framework is particularly suitable for analyzing women's empowerment within households because it allows husbands and wives to have different preferences and views household decisions as the outcome of a bargaining process rather than as the choice of a single unified decision maker. In this setting, household decisions are jointly determined and the influence each spouse exerts depends on their relative bargaining position. A woman's bargaining power is shaped by the resources available to her, her outside options, and the social norms that govern gender relations. Electrification can therefore affect women's empowerment to the extent that it

improves their access to resources, expands their opportunities, or alters the conditions under which household bargaining takes place.

3.2. Potential Channels Linking Electrification to Women's Empowerment.

Electrification may influence women's bargaining position through several interconnected channels, as summarized in Figure 1. By changing the conditions under which women allocate time, access information, move through public space, and participate in household and economic activities, electrification can strengthen women's access to resources and improve their position within the household. These effects are especially relevant in rural settings, where women often face binding constraints arising from domestic responsibilities, limited mobility, weak access to information, and unequal gender norms. The most important channel operates through time use and household production. For instance, women in rural India and other developing countries bear a disproportionate share of responsibility for meeting household energy needs and performing domestic work (Standal and Winther, 2016; Ferrant and Thim, 2019). Access to electricity may encourage a shift away from traditional, time-intensive energy sources and facilitate the use of household technologies, thereby reducing the time and effort required for home production (Dinkelman, 2011). These time savings may allow women to reallocate effort toward paid work, education, childcare, or leisure, all of which can improve their bargaining position within the household.

Another important channel works through improved lighting and perceived safety. Electrification can improve lighting within the home and in public spaces, making it easier for women to remain outside the household after dark for work, schooling, or community activities (Daka and Ballet, 2011; Clancy et al., 2002). In settings where women's mobility is constrained by safety concerns and restrictive norms, better lighting may expand their freedom of movement and public participation. In addition, improved access to electricity may also support educational activities by allowing girls to study at night and by improving conditions for evening schooling, with potentially important consequences for women's long-run autonomy (Kanagawa and Nakata, 2008; Khandker et al., 2009b). Another channel operates through access to household appliances and local economic opportunities. Electrification can increase the adoption of appliances and communication technologies that improve comfort, reduce the physical burden of

domestic tasks, and create opportunities for home-based income generation (Coen-Pirani et al., 2010; Bose et al., 2022). It may also stimulate local economic activity, expanding employment opportunities for women and enabling small-scale enterprises. To the extent that these changes increase women's productivity, earnings potential, or outside options, they may strengthen women's influence over household decisions.

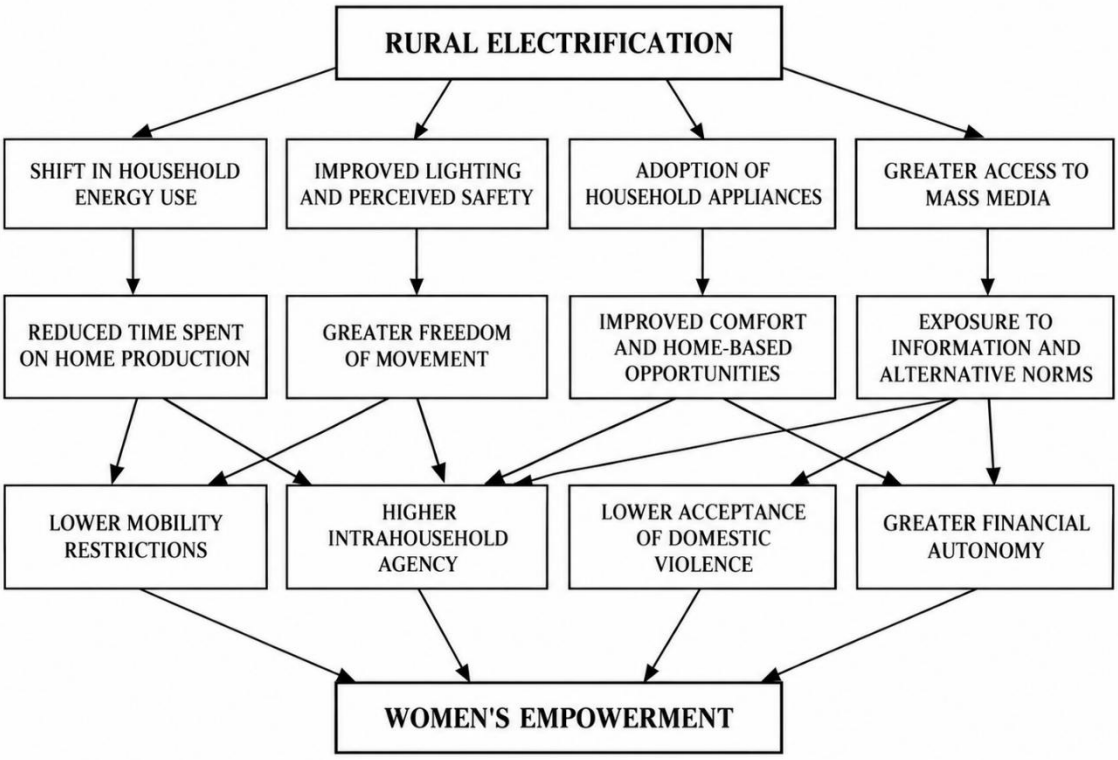
Another important channel relates to health and well-being. Electrification may reduce women's exposure to indoor air pollution by facilitating a move away from kerosene and biomass-based energy sources, which are associated with serious health risks for women who spend more time engaged in cooking and other household activities (Duflo et al., 2008; Parikh, 2011; Khandker et al., 2014). Electrification may also improve the functioning of local health infrastructure by enabling clinics and hospitals to operate at night and use modern equipment (Chaudhury and Hammer, 2004). These improvements may be especially valuable for women, who often face time and mobility constraints in seeking care.

The last and one of the most important drivers relates to electrification influencing women's empowerment by expanding access to information through mass media. Greater access to electricity increases the likelihood that households purchase and use televisions, radios, and other communication technologies, exposing women to information and alternative gender norms (Winther et al., 2017; Nieto, 2025). Existing evidence shows that exposure to mass media can improve women's autonomy, reduce the acceptability of domestic violence and increase political participation (Jensen and Oster, 2009; Okuyama, 2025). Such effects may operate not only within households, but also through shared viewership and the diffusion of ideas within communities. These channels may also shape attitudes toward domestic violence, although the direction of the effect is theoretically ambiguous. On the one hand, better lighting, reduced fuel collection in isolated areas, and stronger bargaining power may reduce women's vulnerability to violence (Guzmán et al., 2020; Rodriguez, 2022). On the other hand, greater mobility and participation in economic or educational activities may expose women to backlash or other forms of risk in contexts with restrictive gender norms (Chakraborty et al., 2018; Mehrotra and Parida, 2017; Mishra et al., 2021). As Kabeer (2014) argues, women's vulnerability is shaped by their position in the local gender hierarchy. For this reason, whether electrification

ultimately reduces women's acceptance of domestic violence remains an empirical question.

Taken together, Figure 1 highlights how electrification may affect women's empowerment through multiple, mutually reinforcing pathways. By easing domestic constraints, improving safety, expanding access to information and household technologies, and enhancing women's opportunities both inside and outside the household, electrification may strengthen women's bargaining position and improve outcomes related to mobility, intrahousehold agency, attitudes toward domestic violence, and financial autonomy.

Figure 1: Theory of Change



4. Data

The data used for the empirical analysis consists of household surveys and census data spanning the period from 2001 to 2012.

4.1 Population Census

I use village-level data from the 2001 and 2011 Population Censuses of India to examine changes in rural electrification during the RGGVY implementation period. The 2001 Census provides information on village population and pre-program electrification status, while the 2011 Census reports dummy indicators for the village's first grid connection in the domestic, agricultural, and commercial sectors. Following Burlig and Preonas (2024), I restrict the sample to single-habitation villages using information from the 2003 and 2009 habitation censuses conducted under the National Rural Drinking Water Programme[4]. This restriction ensures that village population corresponds to the habitation-level population used to determine program eligibility[5]. Using this, I identify single-habitation villages that were completely unelectrified in 2001 and had achieved full electrification across all three sectors by 2011. I then aggregate this information to the district level to construct measures of realized electrification, consistent with the district-level treatment used in the empirical analysis. The 2001 Census also provides district-level socioeconomic characteristics and measures of public goods provision, including overall and female literacy rates, overall and female work participation rates, the number of primary, middle, secondary, and senior secondary schools, hospitals, maternity and child welfare centers, primary health centers, drinking water facilities, and commercial banks. I use these predetermined characteristics to assess the robustness of the results to differences in pre-program district conditions.

4.2 Indian Human Development Survey

For my main analysis, I use the Indian Human Development Survey (IHDS) panel data from 2004-2012, the largest panel survey in India, conducted jointly by researchers from the University of Maryland and the National Council of Applied Economic Research in New Delhi, India. The survey covers a wide range of demographic and socio-economic

[4] The National Rural Drinking Water Programme conducted habitation censuses in 2003 and 2009, which Burlig and Preonas (2024) matched to Census villages using a modified version of the fuzzy-matching algorithm developed by Asher and Novosad (2020). See Burlig and Preonas (2024) for further details on the matching procedure.

[5] Village population and pre-program electrification status determine eligibility under the RGGVY program.

Table 1. Description of Outcome Variables

Category	Survey question	Outcome variable
I. Female mobility	Do you need to ask permission from your husband or a senior family member to visit the local health center? (Yes/No)	Health center (=1 if yes)
	Do you need to ask permission from your husband or a senior family member to visit the homes of relatives or friends in the village? (Yes/No)	Friend home (=1 if yes)
	Do you need to ask permission from your husband or a senior family member to go to the local kirana (grocery) shop? (Yes/No)	Kirana shop (=1 if yes)
	Index constructed by author: (Health center + Friend home + Kirana shop) / 3	**Mobility Restrictions**
II. Norms regarding domestic violence	In your community, is it usual for a husband to beat his wife if she goes out without informing him? (Yes/No)	Leaves without permission (=1 if yes)
	In your community, is it usual for a husband to beat his wife if he suspects her of an extramarital relationship? (Yes/No)	Extramarital affair (=1 if yes)
	In your community, is it usual for a husband to beat his wife if her natal family does not provide the expected dowry in money, jewelry, or other items? (Yes/No)	Insufficient dowry (=1 if yes)
	In your community, is it usual for a husband to beat his wife if she neglects the house or the children? (Yes/No)	House neglect (=1 if yes)
	In your community, is it usual for a husband to beat his wife if she does not cook food properly? (Yes/No)	Poor cooking (=1 if yes)
	Index constructed by author: (Leaves without permission + Extramarital affair + Insufficient dowry + House neglect + Poor cooking) / 5	**Domestic Violence**
III. Intrahousehold agency	Who in your family decides what to cook on a daily basis?	Cooking (=1 if respondent)
	Who in your family decides whether to buy an expensive item such as a TV or refrigerator?	Household purchase (=1 if respondent)
	Who in your family decides how many children you have?	Number of children (=1 if respondent)

Category	Survey question	Outcome variable
	Who in your family decides what to do when a child falls sick?	Child illness (=1 if respondent)
	Who in your family decides how much money to spend on a social function such as a wedding?	Child wedding (=1 if respondent)
	Who in the household took care of the animals in the last 12 months?	Animal care (=1 if respondent)
	Do you and your husband talk about things that happen at work or on the farm? (Yes/No)	Work discussion (=1 if yes)
	Do you and your husband talk about what to spend money on? (Yes/No)	Expenditure discussion (=1 if yes)
	Do you and your husband talk about community matters such as elections or politics? (Yes/No)	Political discussion (=1 if yes)
	Index constructed by author: (Cooking + Household purchase + Number of children + Child illness + Child wedding + Animal care + Work discussion + Expenditure discussion + Political discussion) / 9	**Intrahousehold Agency**
IV. Financial autonomy	Do you personally have any cash in hand to spend on household expenses? (Yes/No)	Cash in hand (=1 if yes)
	Is your name on any bank account? (Yes/No)	Bank account (=1 if yes)
	Is your name on the ownership or rental papers of your home? (Yes/No)	House papers (=1 if yes)
	Index constructed by author: (Cash in hand + Bank account + House papers) / 3	**Financial Autonomy**

Notes: This table shows how the outcome variables were constructed from the Indian Human Development Survey.

characteristics at both the individual and household levels across all states and union territories of India (Desai and Vanneman, 2018).[6] The first survey round was conducted in 2004-05, collecting information on 41,554 rural and urban households. The second

[6] The survey excludes Andaman & Nicobar Islands and Lakshadweep.

round was conducted in 2011-12, covering 42,152 households, with a re-interview rate of 83% of the original households, including split households.[7]

Table 2. Summary Statistics of Outcome Variables

Variable	Obs	Mean	Std. dev.	Mean (2005)	Mean (2011)
Outcome variables (IHDS)					
Health center	30,222	0.79	0.40	0.80	0.79
Friend home	30,066	0.73	0.44	0.78	0.69
Kirana shop	25,043	0.58	0.49	0.59	0.57
Mobility	**24,933**	**0.69**	**0.39**	**0.71**	**0.68**
Leave wo permission	30,276	0.49	0.50	0.43	0.55
Extra-marital affair	30,260	0.87	0.33	0.89	0.86
No dowry	30,270	0.31	0.46	0.28	0.33
House neglect	30,282	0.40	0.49	0.35	0.44
Bad cooking	30,267	0.32	0.47	0.30	0.34
Domestic Violence	**30,203**	**0.48**	**0.33**	**0.45**	**0.50**
Cooking	30,356	0.95	0.22	0.94	0.96
HH purchase	30,260	0.75	0.43	0.69	0.81
N of children	29,592	0.86	0.35	0.79	0.93
Child illness	29,430	0.88	0.32	0.84	0.92
Child wedding	29,189	0.84	0.37	0.78	0.90
Animal care	19,137	0.44	0.50	0.39	0.47
Work discuss	30,377	0.83	0.37	0.82	0.85
Exp discuss	27,386	0.90	0.30	0.89	0.92
Polit discuss	30,377	0.68	0.47	0.64	0.72
Agency	**16,322**	**0.82**	**0.17**	**0.79**	**0.85**
Cash in hand	30,319	0.87	0.34	0.81	0.93
Bank account	30,243	0.49	0.50	0.29	0.69
House paper	29,237	0.16	0.37	0.15	0.18
Financial Autonomy	**29,149**	**0.51**	**0.26**	**0.42**	**0.60**

Note: The IHDS sample consists of married women above the age of 15 living in rural areas.

The distinguishing feature of the IHDS is that it contains detailed information on demographic, socio-economic characteristics and energy use at the household level, along with a separate health and education questionnaire given to one randomly selected eligible women[8] in each household, covering topics related to marriage, gender norms, and health-related investments. To consolidate this information, I merged the household-level data

[7] Split households refer to households that were surveyed during the first round but had divided into multiple households by the time the second round was conducted, often because children had formed their own households and were residing separately.

[8] Eligible women consist of ever-married women aged between 15 and 49.

with the eligible women's data and restrict the sample to rural households, consistent with the focus of RGGVY. I then merge the survey data with the district-level electrification measures constructed from the Population Census. I use the eligible women's questionnaire to construct four main categories of outcome variables: (i) restrictions on women's mobility, (ii) intrahousehold agency, (iii) attitudes toward domestic violence, and (iv) financial autonomy. The analysis follows women across the two survey rounds, with the estimation sample varying according to the availability of information for each outcome. Table 1 provides the definitions of the outcome variables and their corresponding survey questions.

I carefully choose the outcome variables to account for the fact that norms related to female autonomy are deeply rooted and may not shift significantly over a medium period of 5 to 10 years. A large body of previous works shows that these indicators tend to respond to improvements in infrastructure and policies that promote spatial integration. For example, Bhukta et al. (2024) find that higher average hours of electricity at the community level led to greater freedom of movement for women. Rajkhowa and Qaim (2022) show that women with access to mobile phones experience fewer restrictions on their mobility. Sedai et al. (2021c) find that greater reliability in electricity supply enhances women's household agency, reduces mobility restrictions, and also contributes to improvements in general health outcomes. Nandwani and Roychowdhury (2024) demonstrate that India's rural road construction program PMGSY reduced mobility restrictions for women and enhanced their decision-making power. Table 2 reports summary statistics for the main variables used in the analysis and highlights substantial constraints on women's autonomy. Around 69% of women face some form of mobility restriction, while 48% report domestic violence is common the community. Although approximately 82% report some participation in household decision-making, only 51% report some degree of financial autonomy. Taken together, these patterns underscore the multidimensional nature of women's empowerment and the persistence of important constraints across different domains.

5. Empirical Strategy

Ideally, I would like to exploit the program rules using the village population threshold in a regression discontinuity design (RDD) framework. However, the IHDS does not provide village-level identifiers that can be linked to the village-level electrification data. The smallest geographic unit that can be consistently matched across the electrification and IHDS data is at the district level. I therefore aggregate village-level electrification information to the district level and exploit the panel structure of the IHDS data to examine the impact of rural electrification on women's outcomes.

For each outcome of interest Y, I estimate the following regression specification:

$$Y_{idt} = \beta_1 Treat_{dt} + \alpha_i + \gamma_t + \delta X_{idt} + \theta C_{dt} + \varepsilon_{idt}, \quad t \in \{2005, 2011\} \tag{1}$$

Where Y_{idt} is an outcome of interest for individual i living in district d in survey year t. $Treat_{dt}$ is a dummy equal to 1 if district d contains at least one single-habitation village[9] that received its first grid connection during the RGGVY program period, defined as a village that was completely unelectrified in 2001 and achieved full electrification across all sectors by 2011, and 0 otherwise[10]. β_1 captures the differential change in women's outcomes between treated and untreated districts following the electrification rollout. I include individual fixed effects[11], α_i and survey-year fixed effects γ_t. X_{idt} is a vector of time-varying individual and household-level demographic controls, including the respondent's age, household size, annual household consumption expenditure, log of household income and household networks. To account for factors that could affect the electrification rollout, I include a vector of cost-related controls (C_{dt}), including household density and average land gradient to capture differences in the cost of extending electricity infrastructure. I compute standard errors clustered at the district level, consistent with the level at which treatment varies (Bertrand et al., 2004; Cameron and Miller, 2015).

[9] I restrict the sample to single-habitation villages, as sub-village population counts are unavailable. This ensures that village population corresponds to the habitation-level population used to determine RGGVY eligibility, which is not suitable for multi-habitation villages.

[10] $Treat_{dt}$ captures the extensive margin of rural electrification, indicating whether electrification reached previously unelectrified villages within a district.

[11] Since individuals are nested within districts, individual fixed effects also absorb time-invariant district characteristics.

The key empirical challenge to identify the causal effect β_1 is that electricity exposure might be endogenous to women's outcomes through unobserved district characteristics. For instance, districts that received electrification may differ systematically from those that did not in baseline economic development, governance quality, infrastructure endowments, and prevailing gender norms, all of which could independently shape women's empowerment outcomes. This concern is particularly relevant because community-level electrification in India is not random (Khandker et al., 2014). Electrification decisions may be driven by the cost-effectiveness of grid expansion, reflecting unmeasured economic advantages in particular areas (Rathi and Vermaak, 2018), as well as by unmeasured political motivations (Sedai et al., 2021a), thereby generating endogeneity through omitted-variable bias. Consequently, if wealthier or better-governed districts were simultaneously more likely to receive electrification and to experience improvements in women's outcomes for unrelated reasons, simple comparisons between electrified and non-electrified districts would conflate the effect of electrification with these pre-existing differences, yielding biased estimates of the program's effects.

To address these identification concerns, I instrument electricity exposure with predetermined variation in RGGVY eligibility by village population. Specifically, I construct a district-level instrument equal to 1 if the district contained at least one single-habitation village that was unelectrified in 2001 and had a population above the program's 300-person eligibility threshold, and 0 otherwise[12]. The instrument therefore exploits differences across districts in the pre-program distribution of unelectrified villages that were eligible for electrification under RGGVY. The validity of this strategy depends on the population-based eligibility rule. Because program eligibility is contingent on populations from the 2001 Census, four years before RGGVY launched in 2005, this assumption almost certainly holds. Moreover, Burlig and Preonas (2024) show that RGGVY generated a discontinuous increase in electrification around the village population threshold of 300, providing evidence that the eligibility rule meaningfully affected subsequent electricity access.

[12] To the best of my knowledge, no other major Indian social program uses the same population threshold of 300 people as an eligibility criterion.

Nevertheless, although eligibility was formally determined by population, other factors may have influenced which eligible villages ultimately received electrification. For instance, pre-existing infrastructure, local development, or other policy interventions could have affected electrification decisions and may also be related to women's outcomes. The identifying assumption therefore requires that, conditional on the fixed effects and controls, the presence of an eligible village affects women's outcomes through its effect on electrification rather than through these alternative channels. To mitigate this concern, I include individual and survey-year fixed effects, time-varying individual and household controls, and cost-related factors associated with the electrification rollout, including household density and average land gradient. The predetermined nature of the eligibility criterion, together with the discontinuous change in electrification at the population threshold documented by Burlig and Preonas (2024), supports the use of eligibility as an instrument for realized electrification.

Using the IV, I estimate the following two-stage least squares (2SLS):

$$Y_{idt} = \sigma_1 \widehat{Treat}_{dt} + \alpha_i + \gamma_t + \delta X_{idt} + \theta C_{dt} + \varepsilon_{idt} \quad (2a)$$

$$Treat_{dt} = \pi_1 Z_{dt} + \alpha_i + \gamma_t + \delta X_{idt} + \theta C_{dt} + v_{idt} \quad (2b)$$

This specification is identical to the panel model in Equation (5), except that I treat realized electrification as endogenous and instrument it using Z_{dt}. $\widehat{Treat}_{dt}$ denotes the predicted values of electrification obtained from the first stage, capturing variation in realized electrification induced by RGGVY eligibility. Both stages include the same set of control variables and fixed effects as in Equation (5).

6. Results

Instrument validity

The IV strategy in Section 5 requires that predetermined RGGVY eligibility is uncorrelated with unobserved determinants of women's outcomes (exogeneity) and affects these outcomes only through its effect on realized electrification (excludability).

A key concern is that districts containing eligible villages may have differed systematically before the program in characteristics that also predict women's outcomes. For example, eligible districts may have differed in prevailing gender indicators, labor market and human capital characteristics, access to public goods and services and market access and connectivity, each of which could independently influence women's empowerment. To assess whether eligibility is systematically associated with observable pre-program differences along these dimensions, Table 1 reports balance tests examining the relationship between the instrument and a range of predetermined district characteristics.

Using district-level characteristics constructed from the 2001 Population Census, I examine balance across four dimensions. First, eligible districts may have differed in prevailing gender conditions that could independently shape women's empowerment. To capture these differences, I examine pre-program measures of women's socioeconomic status and gender inequality, including female literacy, female labor force participation, and the sex ratio. Second, eligible districts may have differed in labor market and human capital conditions that could affect women's economic opportunities and bargaining position independently of electrification. I therefore examine measures of overall literacy, educational attainment, employment, and the sectoral composition of employment. Third, eligible districts may have differed in their access to public goods and social services, which could contribute to subsequent improvements in women's outcomes. I assess this possibility using pre-program measures of access to schools, health facilities, maternity centers, and child welfare centers. Finally, eligible districts may have differed in market access and connectivity, potentially placing them on different development trajectories even in the absence of electrification. I therefore examine pre-program measures of road and transport connectivity and other available indicators of market access. Table 3 shows that none of these characteristics are significantly correlated with RGGVY eligibility in pairwise comparisons. Taken together, these results provide support for the validity of the instrument.

Table 3: Balance Tests

Panel A:

	Gender Indicators			Labour Market and Human Capital		
	(1) Female Literacy	(2) Female LFPR	(3) Sex Ratio	(4) Literacy Rate	(5) Work Participation Rate	(6) Agriculture Employment Share
Eligbility	-0.017	4.119	0.002	-1.183	1.708	-0.535
	(0.021)	(3.136)	(0.017)	(2.225)	(1.667)	(1.809)
Observations	232	232	232	232	232	232

Panel B:

	Public Goods and Service Provision					Market Access	
	(1) Primary Schools	(2) Middle Schools	(3) Secondary Schools	(4) Primary Health Centers	(5) Maternity and Child Welfare Centers	(6) Distance to Nearest Town	(7) Paved Road
Eligibility	-0.042	-0.003	-0.028	-0.008*	-0.003	3.597	-0.030
	(0.067)	(0.027)	(0.026)	(0.004)	(0.010)	(2.756)	(0.028)
Observations	232	232	232	232	232	232	232

First Stage

Table 2 Panel C reports the first stage estimates from the 2SLS specification. The instrument strongly predicts realized electrification across all outcome samples. The Kleibergen–Paap F-statistics range from approximately 270 to 420, all well above the conventional weak-instrument thresholds (Stock and Yogo, 2002). The first-stage coefficients also align with the RGGVY eligibility rule: districts with villages above the 300-person population threshold were substantially more likely to receive electrification. Across specifications, eligibility increases the probability of treatment by about 82 to 85 percentage points, confirming that the population threshold strongly predicted program rollout.

Impact on mobility restrictions

Although Article 19(1)(d) of the Indian Constitution guarantees the right to free movement, many Indian women still face restrictions. According to IHDS-II data, a large majority need permission from their husbands/senior elder members to go out alone: 78% for health visits, 70% for visiting friends, and 57% even for grocery shopping. Deep-rooted social norms continue to limit their mobility, making freedom of movement a key aspect of women's empowerment.

. I begin by estimating the effect of rural electrification on the mobility restrictions faced by women. Table 4, Column 1 shows that electrification significantly reduces these restrictions. The two-stage least squares (2SLS) estimate in Panel B indicates that electrification induced by RGGVY eligibility reduces mobility restrictions by 13.5 percentage points, statistically significant at the 5% level. These effect sizes are large, corresponding to a 19.2% decline compared to the average. To better understand this effect, appendix Table A1 examines the individual components of the mobility restrictions index. The results show that electrification primarily reduces women's likelihood of having to seek permission to visit friends or local kirana shops, while the effect on seeking permission to access healthcare is smaller and statistically insignificant. These findings suggest that electrification primarily eases restrictions on women's everyday movement outside the household.

Table 4: Impact of Rural Electrification on Women's Empowerment

	(1) Mobility Restrictions	(2) Intra-household Agency	(3) Domestic Violence	(4) Financial Autonomy
Panel A: OLS				
Electrification	-0.071	0.050**	-0.076*	0.025
	(0.045)	(0.024)	(0.041)	(0.024)
Panel B: IV				
Electrification	-0.135**	0.074**	-0.128**	0.036
	(0.054)	(0.030)	(0.050)	(0.030)
Panel C: First Stage				
Eligiblility	0.854***	0.829***	0.853***	0.848***
	(0.043)	(0.050)	(0.042)	(0.043)
Observations	16,955	10,067	23,810	22,333
Districts	184	179	184	184
Mean outcome	0.703	0.758	0.469	0.407
F statistic	394.24	272.18	419.42	392.88

Notes: The table reports estimates of the effect of rural electrification on women's empowerment using the IHDS panel. Panel A reports OLS estimates, Panel B reports two-stage least squares (2SLS) estimates, and Panel C reports the corresponding first-stage estimates. Electrification is an indicator equal to one for districts containing at least one single-habitation village that was completely unelectrified in 2001 and had achieved full grid electrification across the domestic, agricultural, and commercial sectors by 2011. Eligibility is an indicator equal to one for districts containing at least one single-habitation village that was unelectrified in 2001 and had a population above the RGGVY eligibility threshold of 300 persons. All specifications include individual and survey-wave fixed effects, individual and household controls for age, household size, annual household consumption expenditure, log household income, and household networks, as well as a vector of district-level cost controls including household density and average land gradient. The bottom rows report the first-stage coefficient on eligibility and the Kleibergen–Paap first-stage F-statistic. Standard errors, reported in parentheses, are clustered at the district level. See Table 1 for definitions of the outcome variables. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Impact on intra-household agency

Column 2 in Table 4 presents the estimated effect of rural electrification on women's participation in household decision-making. The two-stage least squares (2SLS) estimate in Panel B indicates that electrification induced by RGGVY eligibility increases the intrahousehold agency by 7.4 percentage points, statistically significant at the 5% level. These effect sizes are large, corresponding to a 9.8% increase relative to the average. As shown in appendix Table A2, this improvement is primarily driven by an increase in women's likelihood of making decisions about household purchases, a child's illness, and animal care, as well as their likelihood of participating in discussions about household expenditure. These findings suggest that electrification strengthens women's role in household decision-making, particularly in matters concerning household resources and family responsibilities.

Impact on domestic violence

Column 3 in Table 4 presents the estimated effect of rural electrification on women's attitudes toward domestic violence. The two-stage least squares (2SLS) estimate in Panel B indicates that electrification induced by RGGVY eligibility reduces the domestic violence acceptance by 12.8 percentage points, statistically significant at the 5% level. These effect sizes are large, corresponding to a 27.3% decline relative to the mean. As shown in appendix Table A3, this reduction is primarily driven by a lower likelihood of women considering domestic violence justified when wives leave home without their husbands' permission, bring no dowry into marriage, neglect household responsibilities, or cook poorly. These findings suggest that electrification contributes to changes in social norms by reducing women's acceptance of domestic violence. Given the importance of agency in understanding women's empowerment (Kabeer, 1999), reduced acceptance of domestic violence may reflect broader changes in attitudes toward women's autonomy. Such changes are particularly relevant given the documented links between domestic violence and women's health (Aizer, 2010), employment (Chakraborty et al., 2018), and marriage outcomes (Sarkar, 2024).

Impact on financial autonomy

Finally, column 4 in Table 4 presents the estimated effect of rural electrification on women's financial autonomy. The two-stage least squares (2SLS) estimate in Panel B is

positive but statistically insignificant, indicating no significant effect of electrification induced by RGGVY eligibility on women's financial autonomy. Consistent with this finding, the estimates for the individual components of the financial autonomy index in appendix Table A4 are also statistically insignificant. Taken together, these results suggest that, despite improvements in mobility and intrahousehold agency, electrification alone does not lead to significant changes in women's financial autonomy.

7. Discussion

7.1 Robustness Checks

Robustness Check I: Alternative Treatment and Instrument: Continuous Electrification Exposure

A potential concern with the baseline specification is that the binary treatment and instrument do not capture differences in the intensity of electrification and program eligibility across districts[13]. Districts are classified as treated if electrification reaches at least one previously unelectrified village, regardless of the share of the population benefiting from electrification. Similarly, the binary instrument assigns the same eligibility status to districts containing at least one eligible village, irrespective of the share of the population residing in eligible villages. Consequently, the binary specification overlooks potentially important differences in both realized electrification and eligibility intensity across districts. To address this concern, I construct continuous measures of treatment and instrument using village-level electrification and population data. For instance, the treatment is defined as the share of the district population residing in single-habitation villages that were completely unelectrified in 2001 and achieved full electrification across all sectors by 2011. Similarly, the instrument is defined as the share of the district population residing in single-habitation villages that were completely unelectrified in 2001 and had populations above the RGGVY eligibility threshold of 300 persons. I use this predetermined eligibility measure to instrument for actual electrification exposure. This specification allows me to examine whether the baseline

[13] For example, District A, where 20% of the population resides in villages that received electrification, and District B, where the corresponding share is 95%, are both classified as treated, despite substantial differences in electrification coverage. Similarly, districts where 20% and 95% of the population reside in RGGVY-eligible villages are assigned the same value of one, despite considerable differences in eligibility intensity.

findings hold when accounting for differences in the intensity of electrification across districts.

Table 5: Impact of Rural Electrification on Women's Empowerment (Alternative treatment, IV)

	(1) Mobility Restrictions	(2) Intra-household Agency	(3) Domestic Violence	(4) Financial Autonomy
Panel A: OLS Electrification Share	-0.034*	0.031***	-0.040**	0.015
	(0.019)	(0.010)	(0.017)	(0.009)
Panel B: IV Electrification Share	-0.046**	0.044***	-0.050**	0.006
	(0.023)	(0.013)	(0.020)	(0.012)
Panel C: First Stage Eligible Population Share	0.227***	0.227***	0.233***	0.232***
	(0.017)	(0.018)	(0.016)	(0.017)
Observations	16,955	10,067	23,810	22,333
Districts	184	179	184	184
Mean outcome	0.703	0.758	0.469	0.407
F-statistic	178.98	154.88	206.01	189.39

Notes: The table reports estimates of the effect of rural electrification on women's empowerment using the IHDS panel. Panel A reports OLS estimates, Panel B reports two-stage least squares (2SLS) estimates, and Panel C reports the corresponding first-stage estimates. Electrification Share is defined as the share of the district population residing in single-habitation villages that were completely unelectrified in 2001 and had achieved full grid electrification across the domestic, agricultural, and commercial sectors by 2011. Eligible Population Share is defined as the share of the district population residing in single-habitation villages that were completely unelectrified in 2001 and had populations above the RGGVY eligibility threshold of 300 persons. All specifications include individual and survey-wave fixed effects, individual and household controls for age, household size, annual household consumption expenditure, log household income, and household networks, as well as a vector of district-level cost controls including household density and average land gradient. The bottom rows report the first-stage coefficient on Eligible Population Share and the Kleibergen–Paap first-stage F-statistic. Standard errors, reported in parentheses, are clustered at the district level. See Table 1 for definitions of the outcome variables.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5 reports the results using these alternative measures. The 2SLS estimates in Panel B are consistent with the baseline findings. A 10-percentage-point increase in the share of the district population exposed to electrification reduces the mobility restrictions faced by women by 0.46 percentage points and the domestic violence acceptance by 0.50 percentage points, both statistically significant at the 5% level. Moreover, the same increase in exposure raises the women participation in household

decision making by 0.44 percentage points, statistically significant at the 1% level. In contrast, the estimated effect on financial autonomy remains statistically insignificant.

Panel C reports the corresponding first-stage estimates. Eligible population share strongly predicts realized electrification exposure, with positive and statistically significant coefficients ranging from 0.227 to 0.233 across specifications. These estimates indicate that a 10-percentage-point increase in the share of the district population residing in eligible villages is associated with an approximately 2.3 percentage-point increase in the share actually exposed to electrification. The Kleibergen–Paap F-statistics range from approximately 155 to 206, indicating strong instrument relevance under the continuous specification. Taken together, these results show that the main findings are robust to measuring electrification through population exposure rather than a binary indicator.

Robustness Check II: Alternative Instrument: - Variation Around the Eligibility Threshold

A further concern with the baseline specification is that the binary instrument may capture differences in the extent of pre-program under-electrification across districts rather than variation arising specifically from the RGGVY eligibility threshold. For instance, districts with more unelectrified villages in 2001 were mechanically more likely to contain at least one eligible village. If these districts also experienced different trends in women's outcomes, the estimated effects could partly reflect pre-existing differences rather than electrification induced by program eligibility. To address this concern, I construct an alternative instrument defined as the share of the district population residing in single-habitation villages that were unelectrified in 2001 and had populations just above the RGGVY eligibility threshold (301–500). I also control for the corresponding population share residing in single-habitation villages just below the threshold (100–299) and the total number of initially unelectrified villages in each district. This specification more directly exploits variation around the eligibility threshold while accounting for differences in pre-program under-electrification across districts.

Appendix Table A5 reports the results using this alternative instrument. The 2SLS estimates are consistent in direction with the baseline findings. Electrification reduces mobility restrictions by 35.7 percentage points, statistically significant at the 10% level, and the domestic violence acceptance index by 30.3 percentage points, statistically

significant at the 5% level. It also increases intrahousehold agency by 28.5 percentage points, which is statistically significant at the 1% level. In contrast, the estimated effect on financial autonomy remains statistically insignificant[14]. Taken together, these results indicate that the baseline findings persist when the instrument more closely exploits variation around the RGGVY eligibility threshold while accounting for differences in pre-program under-electrification. This reduces the concern that the baseline results are driven solely by districts' initial electrification deficits.

Robustness Check III: District- and Village-Level Public Goods Controls

A potential concern with the baseline specification is that RGGVY electrification may be correlated with pre-existing differences in local infrastructure and public goods provision that also influence women's empowerment. For instance, districts with greater access to public goods before the program may have experienced different changes in women's outcomes independently of electrification. To address this concern, I re-estimate equations (2a) and (2b) by including interactions between the post-program indicator and a set of predetermined district-level public goods characteristics obtained from the 2001 Population Census[15]. These controls allow for differences in women's outcomes over time associated with initial levels of public service provision. Table A6 reports the results after including these additional controls. The estimates remain consistent with the baseline findings. Electrification reduces mobility restrictions by 11 percentage points and the domestic violence acceptance index by 11.8 percentage points, while increasing intrahousehold agency by 6.8 percentage points. All three estimates are statistically significant at the 10% level. The estimated effect on financial autonomy remains statistically insignificant.

[14] While the point estimates under this specification are somewhat larger in magnitude than the baseline estimates, potentially reflecting a different complier population identified by the narrower instrument, the consistency in sign and statistical significance across the outcomes provides further support for the baseline findings.

[15] The public goods included are the number of primary schools, middle schools, secondary schools, senior secondary schools, hospitals (allopathic), maternity and child welfare centers, primary health centers, availability of a drinking water facility, and the number of commercial banks.

While district-level controls help address some endogeneity concerns, RGGVY was implemented at the village level, raising the possibility that village-level public goods could confound the results. Although equations (2a) and (2b) account for village-specific time-invariant factors, time-varying factors correlated with RGGVY exposure may still bias the estimates. To address this, I use data from the IHDS village module and control for the presence of key public goods and welfare programs, including schools, health centers, widow pension schemes, food security schemes, youth skill programs, employment programs, maternity benefits, and housing support. Table A7 reports the corresponding estimates. The results remain broadly consistent with the baseline findings. Electrification reduces mobility restrictions by 11.1 percentage points and the domestic violence acceptance index by 13.3 percentage points, while increasing intrahousehold agency by 7.1 percentage points. The effects are statistically significant at the 10%, 5%, and 5% levels, respectively. The estimated effect on financial autonomy remains statistically insignificant. Taken together, these results indicate that the baseline findings are robust to controlling predetermined district-level public goods and observable time-varying village characteristics.

7.2 Mechanisms

The first potential mechanism through which electrification may improve women's empowerment is by reducing time spent on home production. For instance, by encouraging households to shift away from traditional, time-intensive energy sources, electrification may generate particularly meaningful time savings in rural settings, where women disproportionately bear responsibility for securing basic household resources (Dinkelman, 2011; Sedai et al., 2021b). To examine this possibility, I investigate changes in household energy use. Table 6, Columns 1 and 2 show that electrification increases the use of electricity as the primary lighting source by 6.9 percentage points and the likelihood that modern fuels become the most common cooking fuel in a village by 5.6 percentage points, statistically significant at the 1% and 10% levels, respectively. Given that fuel collection is a time-intensive activity primarily undertaken by women in India and other developing countries (Ferrant and Thim, 2019), the shift toward modern cooking fuels may reduce the time devoted to home production, allowing women to allocate more time to income-generating activities, caregiving, or leisure. However,

without direct time-use data, I cannot establish whether these changes translate into actual time savings for women.

Table 6: Mechanisms: Effect of Rural Electrification on Intermediate Channels

	(1) Electric Light	(2) Modern Fuel	(3) Fan	(4) Telephone	(5) Sewing Machine	(6) Harass	(7) TV
Panel A: OLS Electrification	0.087***	0.030	0.108***	0.027***	0.022	-0.028***	0.036
	(0.022)	(0.026)	(0.022)	(0.010)	(0.015)	(0.009)	(0.029)
Panel B: IV Electrification	0.069***	0.056*	0.135***	0.033***	0.036**	-0.026**	0.059*
	(0.024)	(0.030)	(0.026)	(0.012)	(0.018)	(0.011)	(0.034)
Panel C: First Stage Eligiblity	0.855***	0.855***	0.855***	0.855***	0.855***	0.856***	0.855***
	(0.041)	(0.041)	(0.041)	(0.041)	(0.041)	(0.041)	(0.041)
Observations	24,519	24,519	24,489	24,461	24,489	24,334	24,519
Districts	184	184	184	184	184	184	184
Mean outcome	0.683	0.060	0.523	0.092	0.191	1.101	0.596
F-statistic	431.68	431.68	431.47	431.64	432.50	439.61	431.68

Notes: The table reports estimates of the effect of rural electrification on potential mechanisms underlying women's empowerment using the IHDS panel. Panel A reports OLS estimates, Panel B reports two-stage least squares (2SLS) estimates, and Panel C reports the corresponding first-stage estimates. Electric Light indicates whether electricity is the primary source of household lighting. Modern Fuel indicates whether modern cooking fuel is the predominant cooking fuel. Fan, Telephone, and Sewing Machine indicate household ownership of the respective assets. Harass is a dummy variable equal to 1 if the household head reports that young girls are frequently harassed in the locality, and 0 otherwise. TV indicates whether the respondent regularly watches television. All specifications include individual and survey-wave fixed effects, individual and household controls for age, household size, annual household consumption expenditure, log household income, and household networks, as well as a vector of district-level cost controls including household density and average land gradient. The bottom rows report the first-stage coefficient and the Kleibergen–Paap first-stage F-statistic. Standard errors, reported in parentheses, are clustered at the district level.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

In addition to changes in household energy use, electrification induced by program eligibility increases access to household appliances that may improve women's living conditions and economic opportunities. Using the IHDS asset module, I construct binary indicators for household ownership of electric fans, telephones, and sewing machines. As shown in Columns (3) – (5) of Table 6, electrification increases the likelihood of owning these assets by 13.5, 3.3, and 3.6 percentage points, respectively, with all three effects statistically significant. These improvements may be particularly

beneficial for women, given the unequal distribution of domestic responsibilities in rural India. For instance, electric fans can improve indoor thermal comfort, potentially reducing heat-related discomfort during household activities. Previous research documents the adverse effects of heat exposure on productivity and well-being (Wargocki et al., 2020; Somanathan et al., 2021; Dasgupta et al., 2025). Such that improved indoor conditions may therefore be especially relevant for women who spend considerable time performing domestic tasks.

Moreover, telephones and sewing machines may also expand women's economic opportunities. Telephones can facilitate communication and access to information without requiring physical mobility, which may be valuable where women's movement is restricted (Aker and Mbiti, 2010). On the other hand, sewing machines, meanwhile, can support home-based income-generating activities, potentially allowing women to combine paid work with domestic responsibilities (Coen-Pirani et al., 2010; Bose et al., 2022). Beyond improving household living conditions and expanding economic opportunities, electrification may also enhance women's perceived safety. To examine this channel, I use an indicator measuring whether respondents report that girls in their locality are frequently harassed. Column 6 of Table 6 shows that electrification significantly reduces reported perceptions of harassment. This finding suggests that access to electricity may improve perceived safety, potentially through better outdoor lighting. Improved perceived safety may, in turn, ease restrictions on women's mobility and facilitate their access to education and employment opportunities (Clancy et al., 2002; ILO, 2018).

Finally, access to mass media is another potential mechanism through which electrification may improve women's empowerment. Greater electricity access may encourage television viewing, exposing women to information and alternative perspectives on gender roles. Previous studies show that exposure to mass media, including cable television and radio, can influence women's empowerment and gender norms (Jensen and Oster, 2009; Okuyama, 2025). Consistent with this evidence, Column 7 of Table 6 shows that electrification increases the likelihood that women regularly watch television by 5.9 percentage points, statistically significant at the 10% level. This finding suggests that greater access to information may contribute to improvements in women's agency and attitudes toward domestic violence.

8. Conclusion

This paper finds that rural electrification improves women's empowerment in India, although its effects vary across dimensions. Electrification reduces mobility restrictions and women's acceptance of justifications for domestic violence while increasing their participation in household decision-making. However, I find no statistically significant improvement in financial autonomy, highlighting the distinction between greater personal autonomy and control over economic resources. The findings also shed light on potential mechanisms. Increased use of modern energy sources and household appliances may ease domestic responsibilities, while improvements in perceived safety and greater exposure to television may expand women's mobility and access to alternative gender norms. These patterns suggest that electrification may influence women's empowerment through several interconnected pathways.

These findings offer important policy implications for India and other developing countries seeking to expand electricity access and reduce gender disparities. While much of the existing literature examines targeted interventions aimed at improving women's access to infrastructure and economic opportunities, this study evaluates the broader implications of a large-scale, government-led rural electrification program in a non-experimental setting. The findings highlight the potential of public infrastructure investment to improve women's mobility, intrahousehold agency, and attitudes toward domestic violence, extending its benefits beyond conventional economic outcomes. These implications are particularly relevant to Sustainable Development Goal 5, which aims to achieve gender equality, and underscore the role that rural electrification can play in addressing gender disparities.

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Appendix

Table A1: Mobility Restrictions — Individual Components

	(1) Health Center	(2) Friend Home	(3) Kirana Shop
Panel A: OLS			
Electrification	-0.017	-0.138***	-0.062
	(0.039)	(0.051)	(0.052)
Panel B: IV			
Electrification	-0.069	-0.187***	-0.126**
	(0.047)	(0.059)	(0.062)
Panel C: First Stage			
Eligibility	0.853***	0.854***	0.853***
	(0.041)	(0.041)	(0.043)
Observations	23,859	23,652	17,115
Districts	184	184	184
Mean outcome	0.803	0.780	0.583

Notes: The table reports estimates of the effect of rural electrification on women's empowerment using the IHDS panel. Panel A reports OLS estimates, Panel B reports two-stage least squares (2SLS) estimates, and Panel C reports the corresponding first-stage estimates. Electrification is an indicator equal to one for districts containing at least one single-habitation village that was completely unelectrified in 2001 and had achieved full grid electrification across the domestic, agricultural, and commercial sectors by 2011. Eligibility is an indicator equal to one for districts containing at least one single-habitation village that was unelectrified in 2001 and had a population above the RGGVY eligibility threshold of 300 persons. All specifications include individual and survey-wave fixed effects, individual and household controls for age, household size, annual household consumption expenditure, log household income, and household networks, as well as a vector of district-level cost controls including household density and average land gradient. The bottom rows report the first-stage coefficient on eligibility and the Kleibergen–Paap first-stage F-statistic. Standard errors, reported in parentheses, are clustered at the district level. See Table 1 for the definition of the mobility restrictions measure and its individual components.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A2: Intrahousehold Agency — Individual Components

	(1) cooking	(2) HH Purchase	(3) No. Child	(4) Child ill	(5) Child Wed.	(6) Disc. Work	(7) Disc. Exp.	(8) Disc. Politics	(9) Animal care
Panel A: OLS									
Electrification	0.012	0.098**	0.032	0.080**	0.039	-0.015	0.043**	-0.070	0.060*
	(0.014)	(0.049)	(0.041)	(0.033)	(0.042)	(0.030)	(0.021)	(0.052)	(0.036)
Panel B: IV									
Electrification	0.014	0.122**	0.022	0.089**	0.064	-0.010	0.051**	-0.064	0.128***
	(0.016)	(0.058)	(0.048)	(0.038)	(0.051)	(0.036)	(0.025)	(0.062)	(0.044)
Panel C: First Stage									
Eligibility	0.85***	0.85***	0.85***	0.85***	0.85***	0.85***	0.85***	0.85***	0.83***
	(0.04)	(0.04)	(0.04)	(0.04)	(0.04)	(0.04)	(0.04)	(0.04)	(0.05)
Observations	24,069	23,887	23,034	22,687	22,301	24,099	24,099	24,099	11,231
Districts	184	184	184	184	184	184	184	184	180
Mean outcome	0.948	0.697	0.772	0.840	0.769	0.823	0.891	0.665	0.368

Notes: The table reports estimates of the effect of rural electrification on women's empowerment using the IHDS panel. Panel A reports OLS estimates, Panel B reports two-stage least squares (2SLS) estimates, and Panel C reports the corresponding first-stage estimates. Electrification is an indicator equal to one for districts containing at least one single-habitation village that was completely unelectrified in 2001 and had achieved full grid electrification across the domestic, agricultural, and commercial sectors by 2011. Eligibility is an indicator equal to one for districts containing at least one single-habitation village that was unelectrified in 2001 and had a population above the RGGVY eligibility threshold of 300 persons. All specifications include individual and survey-wave fixed effects, individual and household controls for age, household size, annual household consumption expenditure, log household income, and household networks, as well as a vector of district-level cost controls including household density and average land gradient. The bottom rows report the first-stage coefficient on eligibility and the Kleibergen–Paap first-stage F-statistic. Standard errors, reported in parentheses, are clustered at the district level. See Table 1 for the definition of the intrahousehold agency measure and its individual components.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A3: Domestic Violence— Individual Components

	(1) Leave w/o permission	(2) Beat if extramarital	(3) Beat if no dowry	(4) Beat if house neglect	(5) Beat if bad cooking
Panel A: OLS					
Electrification	-0.157***	0.018	-0.053	-0.092	-0.102*
	(0.057)	(0.039)	(0.050)	(0.056)	(0.053)
Panel B: IV					
Electrification	-0.209***	0.022	-0.107*	-0.191***	-0.162**
	(0.070)	(0.047)	(0.061)	(0.068)	(0.065)
Panel C: First Stage					
Eligibility	0.853***	0.851***	0.852***	0.852***	0.852***
	(0.042)	(0.042)	(0.042)	(0.042)	(0.042)
Observations	23,938	23,897	23,923	23,951	23,917
Districts	184	184	184	184	184
Mean outcome	0.444	0.894	0.306	0.377	0.326

Notes: The table reports estimates of the effect of rural electrification on women's empowerment using the IHDS panel. Panel A reports OLS estimates, Panel B reports two-stage least squares (2SLS) estimates, and Panel C reports the corresponding first-stage estimates. Electrification is an indicator equal to one for districts containing at least one single-habitation village that was completely unelectrified in 2001 and had achieved full grid electrification across the domestic, agricultural, and commercial sectors by 2011. Eligibility is an indicator equal to one for districts containing at least one single-habitation village that was unelectrified in 2001 and had a population above the RGGVY eligibility threshold of 300 persons. All specifications include individual and survey-wave fixed effects, individual and household controls for age, household size, annual household consumption expenditure, log household income, and household networks, as well as a vector of district-level cost controls including household density and average land gradient. The bottom rows report the first-stage coefficient on eligibility and the Kleibergen–Paap first-stage F-statistic. Standard errors, reported in parentheses, are clustered at the district level. See Table 1 for the definition of the domestic violence measure and its individual components.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A4: Financial Autonomy— Individual Components

	(1) Cash in hand	(2) Bank Account	(3) House Paper
Panel A: OLS			
Electrification	0.065**	-0.002	0.002
	(0.031)	(0.038)	(0.035)
Panel B: IV			
Electrification	0.100***	0.006	-0.006
	(0.037)	(0.045)	(0.046)
Panel C: First Stage			
Eligibility	0.852***	0.852***	0.848***
	(0.042)	(0.042)	(0.043)
Observations	24,007	23,863	22,481
Districts	184	184	184
Mean outcome	0.799	0.267	0.149

Notes: The table reports estimates of the effect of rural electrification on women's empowerment using the IHDS panel. Panel A reports OLS estimates, Panel B reports two-stage least squares (2SLS) estimates, and Panel C reports the corresponding first-stage estimates. Electrification is an indicator equal to one for districts containing at least one single-habitation village that was completely unelectrified in 2001 and had achieved full grid electrification across the domestic, agricultural, and commercial sectors by 2011. Eligibility is an indicator equal to one for districts containing at least one single-habitation village that was unelectrified in 2001 and had a population above the RGGVY eligibility threshold of 300 persons. All specifications include individual and survey-wave fixed effects, individual and household controls for age, household size, annual household consumption expenditure, log household income, and household networks, as well as a vector of district-level cost controls including household density and average land gradient. The bottom rows report the first-stage coefficient on eligibility and the Kleibergen–Paap first-stage F-statistic. Standard errors, reported in parentheses, are clustered at the district level. See Table 1 for the definition of the financial autonomy measure and its individual components.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A5: Robustness to an Alternative Instrument Based on the RGGVY Eligibility Threshold

	(1) Mobility Restrictions	(2) Intra-household Agency	(3) Domestic Violence	(4) Financial Autonomy
Electrification	-0.357*	0.285***	-0.303**	0.103
	(0.191)	(0.100)	(0.148)	(0.081)
Observations	16,955	10,067	23,810	22,333
Districts	184	179	184	184
Mean outcome	0.703	0.758	0.469	0.407
F-statistic	18.76	17.72	18.07	16.42

Notes: The table reports two-stage least squares (2SLS) estimates of the effect of rural electrification on women's empowerment using an alternative instrument based more closely on the RGGVY population eligibility threshold. Electrification is defined as in the baseline specification. The alternative instrument is the share of initially unelectrified villages in the district with populations just above the 300-person eligibility threshold (301–499), interacted with the post-program indicator. To account for differences in the overall extent of pre-program under-electrification across districts, I control for the corresponding share of initially unelectrified villages with populations just below the threshold (100–299) and the total number of initially unelectrified villages in the district, both interacted with the post-program indicator. This specification therefore exploits variation in whether initially unelectrified villages fell just above rather than below the RGGVY eligibility threshold. All specifications include individual and survey-wave fixed effects, individual and household controls for age, household size, annual household consumption expenditure, log household income, and household networks, as well as district-level cost controls for household density and average land gradient. The bottom row reports the Kleibergen–Paap first-stage F-statistic. Standard errors, reported in parentheses, are clustered at the district level. See Table 1 for definitions of the outcome variables.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A6: Robustness to District-Level Public Goods Controls

	(1) Mobility Restrictions	(2) Intra-household Agency	(3) Domestic Violence	(4) Financial Autonomy
Electrification	-0.110*	0.068*	-0.118*	0.037
	(0.063)	(0.038)	(0.060)	(0.035)
Observations	17,246	10,244	24,222	22,710
Districts	184	179	184	184
Mean outcome	0.704	0.759	0.470	0.407
F-statistic	267.54	195.32	278.06	256.79

Notes: The table reports two-stage least squares (2SLS) estimates of the effect of rural electrification on women's empowerment after interacting a set of predetermined district-level public goods controls from the 2001 Population Census with the post-program indicator. These controls include the number of primary, middle, secondary, and senior secondary schools; hospitals; maternity and child welfare centers; primary health centers; the presence of a drinking water facility; and the presence of a commercial bank. Each public goods measure is interacted with the post-program indicator, allowing outcomes to evolve differently according to pre-program levels of public service provision. Electrification is defined as in the baseline specification and is instrumented using predetermined RGGVY eligibility. All specifications include individual and survey-wave fixed effects, individual and household controls for age, household size, annual household consumption expenditure, log household income, and household networks, as well as district-level cost controls for household density and average land gradient. The bottom row reports the Kleibergen–Paap first-stage F-statistic. Standard errors, reported in parentheses, are clustered at the district level. See Table 1 for definitions of the outcome variables.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A7: Robustness to Time-Varying Village-Level Controls

	(1) Mobility Restrictions	(2) Intra-household Agency	(3) Domestic Violence	(4) Financial Autonomy
Electrification	-0.111*	0.071**	-0.133**	0.020
	(0.057)	(0.032)	(0.054)	(0.030)
Observations	15,528	9,268	22,049	20,661
Districts	171	167	171	171
Mean of dep. variable (2004-05)	0.701	0.759	0.476	0.410
Kleibergen-Paap F-statistic	331.59	217.15	341.11	318.94

Notes: The table reports two-stage least squares (2SLS) estimates of the effect of rural electrification on women's empowerment after including time-varying village-level public goods and welfare program controls from the IHDS village modules. These controls capture the availability of schools and health centers, as well as whether the village has an active widow pension scheme, food security program, youth education and skill-development program, employment program, maternity benefit scheme, housing program, safe drinking water, and sanitation facilities. Electrification is defined as in the baseline specification and is instrumented using predetermined RGGVY eligibility. All specifications include individual and survey-wave fixed effects, individual and household controls for age, household size, annual household consumption expenditure, log household income, and household networks, as well as district-level cost controls for household density and average land gradient. The bottom row reports the Kleibergen–Paap first-stage F-statistic. Standard errors, reported in parentheses, are clustered at the district level. See Table 1 for definitions of the outcome variables.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$